PROJECT REPORT

EV Market Performance & Insights

Electric vehicle sales, battery technology and driving range analytics

Project	Details
Prepared by	Dhruv Shrivastava
Institute	Medicaps University, Indore
Project type	Student analytics project
Dataset	200,000 vehicle records; 37 source fields
Deliverables	Public website, Tableau workbook and story, prepared data, project report

A guided analysis of recorded electric vehicle market indicators through interactive visualizations.

Live dashboard: https://zelusgit.github.io/ev-market-performance-insights-complete/

PROJECT OVERVIEW & LINKS

This report describes the project scope, data preparation, dashboard, Tableau story, publication status, and remaining evidence. The public website is deployed on GitHub Pages; the demo video will be linked after recording and Drive access is verified.

Resource	Project link
GitHub repository	Open complete repository
Live dashboard	Open public GitHub Pages site
Tableau Public dashboard and story	Open interactive five-point story
Tableau workbook file	Download packaged workbook
Project report PDF	Open this report in GitHub
Prepared data extract	Download compressed CSV
Story sequence	Read story notes
Demo video	Pending recording and verified Google Drive link

Interpretation note

Values describe the supplied dataset and should not be treated as audited market totals or forecasts. Annual sales and customer ratings include missing entries; each measure excludes blank values for that calculation. The source file itself is not redistributed; only the prepared extract is included.

1. INTRODUCTION

EV Market Performance & Insights is a student analytics project that presents recorded electric vehicle indicators through a browser dashboard and a Tableau story. Users can compare sales by manufacturer and category, inspect battery technology mix, and review driving range by vehicle category.

1.1 Purpose and scope

The project organizes selected measures into summary cards, charts, filters, and a guided five-point story. It covers the supplied 200,000-row training dataset and ten fields selected for the dashboard. It is descriptive and exploratory; it does not establish market representativeness, causal relationships, or forecast accuracy.

2. IDEATION PHASE

Vehicle sales, battery types, and range can be difficult to compare across a large flat dataset. The dashboard groups these measures to support exploration, while the Tableau story guides viewers from an overview to focused comparisons.

User need	Project response
Understand the dashboard quickly	Overview KPIs and concise chart titles
Compare market segments	Category, manufacturer, country and model-year filters
Explore technology choice	Battery type distribution view
Review a guided narrative	Five sequential Tableau story points

3. REQUIREMENT ANALYSIS

Stage	User action	Expected experience
Discover	Open the public project site	Understand purpose and available views
Explore	Choose filters	Focus the comparisons
Interpret	Review charts and story	Compare recorded patterns
Share	Open project resources	Access source, workbook, report, and data

3.1 Solution requirements

Responsive dashboard with KPI cards and filters; views for category, year, manufacturers, market share, battery mix, range and country; a prepared dataset with documented missing-value handling; a Tableau workbook with five story points; and project documentation with live links and completion status.

3.2 Data flow and technology

Input	Processing	Output
Supplied EV training data	Select ten fields; trim text; preserve blanks	Compressed prepared extract and Tableau data source
Prepared records	Filter and aggregate selected measures	KPIs, charts and guided story

Technology	Role
HTML, CSS, JavaScript	Static browser dashboard and interactions
Gzip CSV	Prepared data extract loaded in browser
Tableau Public workbook	Packaged analytical views and story
GitHub and GitHub Pages	Project repository and published site

4. PROJECT DESIGN

Summary cards support quick scanning, filters support focused exploration, and the story sequence moves from overview to manufacturer performance, market share, battery technology, and range. The website is published at the live Pages URL in the links section.

4.1 Proposed solution

A standalone static website with five filters and interactive market views; a Tableau workbook with five story points and descriptive captions; a prepared extract retaining all 200,000 rows and ten selected fields; and project documentation describing data decisions and remaining steps.

4.2 Project plan and current evidence

Phase	Work item	Deliverable / status
Define	Set scope and comparison questions	Project objectives
Prepare	Inspect fields, blanks and duplicate IDs	Prepared dataset and notes
Build	Create dashboard and views	HTML, CSS and JavaScript
Narrative	Arrange Tableau sequence	Five captions in published workbook
Document	Describe data, site, story and status	This report and repository notes
Publish	Deploy site and record demo	Pages and Tableau Public are live; video is pending

The prepared data extract is downloadable separately. View the interactive Tableau dashboard and five-point story using the Tableau Public link above; the downloadable workbook file is also available.

5. FUNCTIONAL AND PERFORMANCE CHECKS

Test area	Check	Status
Website load	Open hosted page; confirm dataset and dashboard appear	Site published; data loading verified earlier
Filters and reset	Change filters and restore defaults	Functional plan; detailed results not recorded
Story sequence	Navigate all five points	Five-point story is in packaged workbook
Responsive layout	Check desktop and narrow viewport	Pending measurement
Performance	Measure load, filtering and chart update	Pending measurement

5.1 Performance testing

No measured browser timings or stress-test results are available. Record compressed data load, filter response, chart update, and narrow-screen behavior in the performance notes with a named device/browser and repeatable method.

6. RESULTS

The project includes a static website dashboard, a compressed prepared CSV containing 200,000 rows and ten selected fields, a packaged Tableau workbook, and this project report.

Point	Story focus
1	EV Market Overview
2	Manufacturer Sales Performance
3	Market Share Across Manufacturers
4	Battery Technology Preferences
5	Driving Range by Vehicle Category

6.1 Data preparation summary

The source contains 200,000 records and 37 fields. Ten dashboard fields were retained, headers were made readable, and surrounding whitespace was trimmed. Blank values were preserved: 2,028 annual sales values and 3,977 customer ratings are blank. There are 199,999 distinct vehicle IDs, so one identifier repeats; both rows remain in the extract.

7. ADVANTAGES, LIMITATIONS & NEXT STEPS

Advantages. The project combines sales, battery mix, and range comparisons in a single exploratory interface; uses a prepared compressed extract and static website; presents a five-part Tableau narrative; and documents missing-value handling.

Limitations. The dataset's representativeness and metric definitions have not been independently validated. Annual sales may contain extreme values that should be reviewed. The dashboard counts source rows and does not equate row count to distinct vehicles. Performance measures and narrow-screen results remain unverified. The demo video is pending recording.

7.1 Conclusion

The project organizes a large EV dataset into a public website dashboard and guided Tableau story. It gives reviewers a practical way to compare recorded sales, battery technology, and driving range while preserving caveats about data quality and interpretation. The website and Tableau story are ready for review; the video link will be added after recording and sharing.

7.2 Next steps

Review metric outliers and source definitions; record browser performance and responsive-layout results; record the project walkthrough; and add the verified Google Drive video link to the repository, PDF, website, and SkillWallet portal.

DATA NOTES

The original *ev_market_train.csv* stays local; the repository contains only the prepared extract at *data/cleaned/ev_market_cleaned.csv.gz*. The extract has 2,028 blank sales values, 3,977 blank ratings, and one repeated vehicle ID. See the linked data preparation notes for details.

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